Building maplibre-terrain

A walkthrough of designing, building, and shipping a keyless 3D terrain viewer with MapLibre GL JS

Repository: github.com/rsignell/maplibre-terrain

Live app: rsignell.github.io/maplibre-terrain/terrain/

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> This document is a teaching artifact: a readable transcript of the conversation that produced `maplibre-terrain`, from the first idea ("can I view 3D US topography without an API key?") through a working, deployed, mobile-friendly map viewer. It is meant to show the back-and-forth flow of building something with Claude Code — not just the final code.

Part 1 — Scoping the idea

RICH

I would like to create a json for geolibre that would allow me to view topography in 3d for the US draped with mapbox imagery and OSM cities and roads, etc. Is that possible?

CLAUDE

Yes — this is a standard MapLibre GL JS style: a `raster-dem` source for elevation driving the `terrain` property, a raster imagery layer draped on top, and an OpenStreetMap-derived vector layer for roads/places/labels. The catch is that the obvious imagery choice (Mapbox satellite tiles) requires an API key.

RICH

Are there sources that don't need api keys?

CLAUDE

Three keyless pieces fit together well:

- Elevation — AWS/Mapzen Terrarium raster-dem tiles (public S3 bucket, no key)
- Imagery — Esri World Imagery (public ArcGIS REST tile service, no key)
- Roads/places/labels — OpenFreeMap vector tiles (OpenMapTiles schema, free hosted TileJSON, no key)

RICH

yes please

CLAUDE

Built `us-3d-terrain.json`: a MapLibre style with the three sources above, a `terrain` block (`exaggeration: 1.5`), a sky layer, and a curated set of OpenFreeMap "Liberty" layers (transportation, place labels, boundaries) with land-use and building fills dropped so the satellite imagery shows through cleanly underneath the roads and labels.

Part 2 — Getting it on the web

RICH

So we can have a URL that allows us to explore 3d terrain in a browser if we push the json to github?

CLAUDE

Yes — a small `index.html` that loads MapLibre GL JS from a CDN and points at the style JSON, served via GitHub Pages, gives a shareable URL with no backend at all.

RICH

Can you enable pages and push what we need

CLAUDE

Enabled GitHub Pages via `gh api repos/rsignell/maplibre-terrain/pages` (serving from `main`, root), committed the style JSON and viewer, and pushed. First live, keyless 3D terrain map, confirmed in the browser.

RICH

this is using cesium, right?

CLAUDE

No — this is MapLibre GL JS, not CesiumJS. MapLibre is a 2.5D WebGL map renderer that drapes a raster-dem elevation grid under the map, tilted with pitch/bearing; it's much lighter-weight than a full 3D globe engine like Cesium, which is a better fit here since we only need regional terrain, not a whole-earth globe.

Part 3 — Search and exaggeration

RICH

Can we add a search bar to zoom to a specific place? Also can we call this geolibre/terrain instead of just geolibre?

CLAUDE

Added a debounced place search box backed by the free, keyless Photon geocoder (komoot), with results flying the map to the matched place via `fitBounds` / `flyTo`. Moved the viewer to a `terrain/` subpath so the repo root stays a simple index of future demos.

RICH

Would it be possible to add a vertical exaggeration widget?

CLAUDE

Added a slider wired to `map.setTerrain({source, exaggeration})`, letting the vertical relief be scaled up or down live.

RICH

let's push it

CLAUDE

Committed and pushed. Confirmed the deployed build via `gh api repos/.../pages/builds/latest` and a direct `curl` of the live page.

Part 4 — Cleaning up the UI

RICH

hmm, I don't see it in the github pages, even after refresh

CLAUDE

The build had succeeded and the new markup was being served — the symptom matched browser caching (`cache-control: max-age=600`) rather than a failed deploy. A hard refresh resolved it.

RICH

Let's hide the info stuff on a simple "i" icon beneath the terrain icon -- it takes up too much space on mobile. Actually, let's have the terrain icon open a menu that allows turning off the vertical exaggeration or scaling it from 1-5.

CLAUDE

Replaced the always-visible slider and attribution bar with two custom MapLibre controls: a terrain icon that opens a small popover menu (on/off toggle + 1–5× exaggeration slider), and a separate, compact info ("i") icon for attribution.

RICH

need to put a thin white circle around the black "i" icon to make it visible on light or dark backgrounds — or make the "i" white -- perhaps it is currently transparent?

CLAUDE

The built-in MapLibre attribution button rendered as a low-contrast solid dot. Replaced it with a custom inline SVG icon — white circle, dark stroke, dark glyph — verified legible on both light and dark map backgrounds.

RICH

Let's add a bit more info about how this app was constructed when the info icon is clicked (needs to scroll and not take up too much real estate on mobile)

CLAUDE

Added a "How this was built" section to the info popover, constrained with `max-height: min(70vh, 420px); overflow-y: auto;` and a mobile-safe max-width, verified at both full-height and short landscape phone viewports.

Part 5 — Sharing a view

RICH

could we add a widget that would allow sharing a URL to capture the current view?

CLAUDE

The map was already created with `hash: true`, so the URL already encodes zoom/lat/lon/bearing/pitch as you navigate. Added a share icon that uses the Web Share API where available, falling back to copying the current URL to the clipboard, with a brief "Link copied" toast.

RICH

what is the url to the repo

CLAUDE

github.com/rsignell/maplibre-terrain

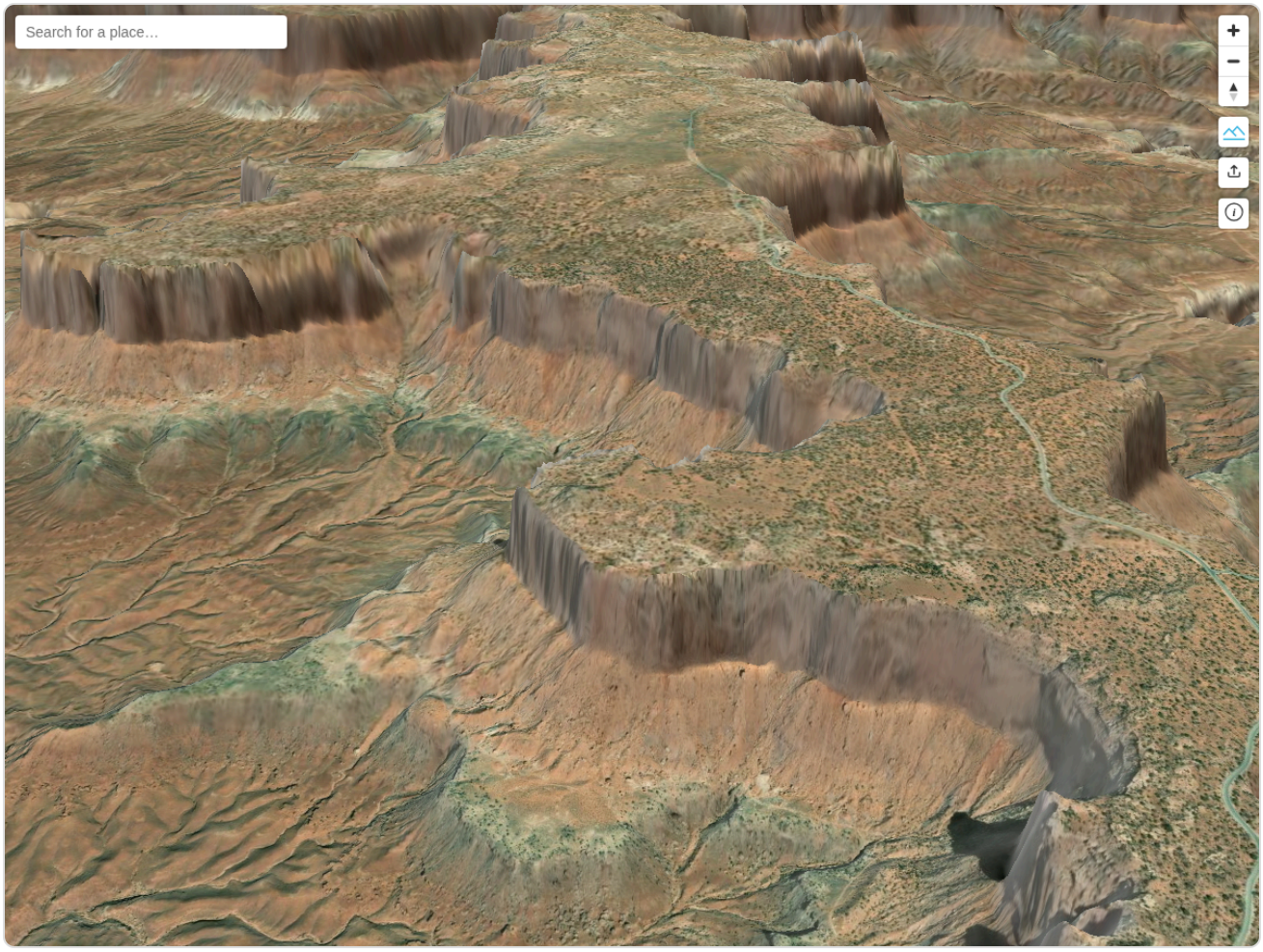
RICH

rename to maplibre-terrain

CLAUDE

Renamed the repository from `geolibre` to `maplibre-terrain` to avoid confusion with an unrelated, much larger open-source project of a similar name. GitHub Pages and the local git remote both followed the rename automatically; updated the one in-page link that still pointed at the old name.

Appendix — Island in the Sky, Canyonlands National Park



Captured from rsignell.github.io/maplibre-terrain/terrain/#14.53/38.32812/-109.8662/0/60